Architectural Particular Specification

CARPENTRY, LAMINATE AND JOINERY

SECTION 5.1

Rev	Date	Description
0	14 APR 2022	First Issue

SECTION 5.1

CARPENTRY, LAMINATE AND JOINERY

PART A - GENERAL

1.01 SUMMARY OF WORK

A. This Section includes, but not limited to the following:
   a. Untreated wood:
      i. Wood framing;
      ii. Wood blocking and nailers;
      iii. Wood furring and grounds;
      iv. Wood sleepers;
      v. Plywood backing panels;
   b. Preservative treated wood, same items;
   c. Fire retardant treated wood, same items;
   d. Casework and cabinetry;
   e. Shelving and ancillary supports; and
   f. Finish Carpentry Hardware, Adhesives and Joinery Fixings.

B. The contractor is required to supply and install cabinets, work counters, information counters and all built-in furniture / shelves / lockers / other carpentries as indicated in drawings.

C. The works shall include the fabrication, transportation, assembly and installation in full conforming to the contract requirements.

D. For cabinets, all necessary hardware and accessories shall be basically of proprietary product to be assembled on site. Each cabinet shall consist of:-
   a. Concrete plinths for base units shall be lined with matching tile skirting.
   b. Wall hung cabinet with back panels, adjustable shelves, handles, lighting pelmet, & doors;
   c. Sink units with work-top, doors, back panels, drawers, handles, splash back, bullnose front edge, adjustable shelves, swivel tray, sit-on sink, mixer taps with angle valves flange and associated fittings, overflow units, waste with plug, chain and stay complete with bottle traps angle valves flange and associated fittings;
   d. Fixings, brackets, screws, beadings, gasket and other accessories;
   e. Sealant compound for filling the gaps around the sink boards & mixer taps and gaps between the cabinets & the wall tile surface;
   f. Forming of opening on counter top of sink unit for placing of sink. Cost for placing of sink shall be deemed to be included in the tender sum;
   g. Forming of openings on drain board or counter top for placing of sink mixer taps. Costs of placing of sink mixer taps shall be deemed to be included in the tender sum;
   h. Forming of openings on the back panels for piping works;
   i. Connection of lightings to power supply up to electrical provisions as specified;
   j. Connection of water supply and drain pipe up to plumbing and drainage provision;
   k. Overshelves with spotlights and fluorescent lights.

E. The Contractor shall also be responsible for providing the following :

a. Shop drawings, catalogues, technical data conforming to specified requirements;
b. Scheduling and monitoring of the work;
c. Samples and mock-ups;
d. Loading test of the wall cabinets and drawers showing the structural adequacy;
e. Co-ordination with work of other trades concerned for the installation of apparatus cabinet appliances and utilities meters;
f. Protection;
g. All final cleaning of the kitchen cabinets and other installation works of this Contract;
h. Coordination with the electrical sub-contractor to ensure the adequacy of power points and the setting out of these power points fit for connecting to the future installation of electrical appliances.

1.02 REFERENCE

(NOT USED)

1.03 QUALITY ASSURANCE

A. The work of this section shall be provided by a firm having a minimum of 5 years experience on projects of similar size and quality to that specified and shown.

B. Single source manufacturing and installation responsibility: engage a qualified manufacturer to assume undivided responsibility for architectural woodwork specified herein, including fabrication, finishing, and installation.

C. Qualification of wood veneer and joinery wood specialist supplier: company specializing in the supply and exporting of tropical hardwood and exotic wood veneer with minimum of ten (10) years of documented experience. The supplier shall have a thorough knowledge of tropical hardwood and exotic wood and their manufacturing and quality.

D. The plastic laminated shall manufactured in strict accordance with is9001 and iso 14001 criteria, and meet or exceed the performance requirements of nema ld3 / bs en438 / iso 4586-2 for high pressure decorative laminates. It also meets the bs 476.

1.04 PERFORMANCE REQUIREMENTS

A. Unless otherwise specified, all materials shall be of the best quality of the type specified.

B. All timber shall be as follows : -

a. High quality,
b. Well and properly seasoned,
c. Mature growth,
d. Free from wood wasp holes, large loose or dead knots over 25mm diameter or other defects,
e. Free from splits or other defects that will reduce its strength,
f. Sawn die-square.

C. All scantlings and boards shall be supplied in such lengths, breadths and thickness as may be required.

D. Sawn timbers must hold the full dimensions specified.

E. Timber shall be cut to the required sizes and lengths.

F. Timber shall be stored in a dry, well ventilated place and protected from weather.

G. The whole of the work shall be framed and finished in accordance with the Standard Details.

H. Work where required shall be fitted with all necessary galvanized mild steel ties, straps, bolts, screws, etc.

I. Surfaces of all joinery works shall be finished with fine abrasive paper unless otherwise specified.

J. Defects due to faulty materials or workmanship shall be made good or replaced at the Contractor's own expense.

K. Samples of timber for joinery shall be submitted for approval where the grain is to be left exposed or where the surfaces are to be varnished or similarly treated.

L. All timber for exposed use shall be equal in appearance and quality to the approved samples.

M. Moisture Content: -

a. Moisture content shall be calculated at the time of fabrication by the following formula :-

$$\frac{\text{Wet (or supplied) Mass} - \text{Dry Mass}}{\text{Dry Mass}} \times 100 = \text{Moisture Content Per Cent}$$

The dry mass shall be determined by drying in an oven at a temperature of 103ºC ±2ºC until the weight is constant.

b. The maximum permissible moisture content in timbers to be incorporated into the work shall be as follows:

i. Internal timber for use in Air Conditioned space: 12%

ii. Internal timber generally: 16%

iii. Timbers with one face to the exterior of the building and one face to the interior (e.g. window frames, doors): 18%

iv. External timbers (e.g. fencing etc.): 20%

c. The Architect may require the contractor to submit laboratory test reports or manufacturer's certificates regarding the moisture of the timber demonstrating compliance of this specification prior to acceptance of the timber materials for the work.

If these percentages cannot be attained due to local circumstances of supply and availability the Architect's attention must be drawn to the fact.

N. Fire Retardant Woodwork:

Provide finished wall and ceiling panels and other woodwork with a Class 0/Class 1 flame spread performance in accordance with the Building Regulations definition and BS 476, Part 7 test standards. Chemical fire-retardant process for wood members shall conform with British Wood Preserving and Damp-proofing Association (BWP) recommendations for pressure treatment, including amount of chemical retention required for the wood species and intended use in the Work. Provide only wood treatment that does not form surface deposits otherwise adversely affect finishes applied to treated wood surfaces. Treat those items required by Building Code to be treated and those items shown or specified as "Fire Retardant Treated Wood". Obtain each type of fire-retardant-treated wood products from one source for both treatment and fire-retardant formulation.

O. Laminates

Plastic laminate shall comply with the following standard:
High Pressure Laminates : Class D3, Fire Rating F1 and Performance Rating HD-Heavy Duty conforming to BS 4965 : 1991.: Color, pattern, and finish selected by Architect from Manufacturer's standard products.

a. Exposed Horizontal Surface : HD grade per BS EN 438 : Part 1
b. Exposed Vertical Surfaces : HG grade per BS EN 438 : Part 1
c. Adhesive : Thermosetting Resorcinol formaldehyde Type WBP adhesive recommended by Manufacturer for D3 and D4 class plastic laminates, complying with BS4965 and BS EN 204.
d. Minimum 1.5 mm thickness. Use D4 class plastic laminate for application with frequent prolonged exposure to running or condensed water.
e. Use plastic laminates conforming to U.S. National Sanitation Foundation NSF #35 requirement for plastic laminate surfaces in contact with foodstuff.
f. Adhesion of laminate to backing board material shall not exhibit any failure of glueline such that any area of either the laminate or the backing board greater then $40mm^2$ is free from adherent matter derived from either backing board or laminate.

P. Workmanship:-

Refer also ASD GS 2012 clauses 13.31.1 to 13.43, 13.51 to 13.59, and 13.64 to 13.66.

a. The Contractor shall design, engineer, test, fabricate, deliver, install and guarantee all construction necessary to provide a complete kitchen and pantry cabinet system, included any measures that may be required to that end, notwithstanding any omissions or inadequacies of drawings and/or specifications. All parts supplied shall be pre-drilled and pre-fabricated where possible and well wrapped and protected against scratching or damage during delivery and fixing.
b. The Contractor shall allow for attendance to cutting or forming holes, openings etc., to cabinet to suit the installation of all services pipe and gas meter inside the cabinets. These holes / openings shall be well sealed. All cutting and other modification work for complete installation shall be deemed to be included in the tender sum.
c. Shop drawings catalogues and technical data / information of the pantry cabinets, together with the details of fitting shall be submitted.

d. The tender drawings indicate only the outline of pantry cabinets required. It is the responsibility of the Contractor to verify at the building site all dimensions shown on the Architect's drawings and incorporate the verified dimensions into their final set of shop drawings before manufacture.

e. The Contractor shall allow for all necessary adjustment of the pantry cabinets to meet actual building site conditions to ensure that the pantry cabinets are erected plumb and true.

f. All lippings should be finished in factory. If lippings have to be done on site to suit actual site condition, the gluing, material and application method shall be subject to the Architect's confirmation of no adverse comment.

g. The Contractor shall place the composite sinks onto the sink unit. Sealing to surround to sink by silicone sealant to be included. Silicone sealant to be submitted to the Architect for comment.

h. The Contractor shall provide all necessary fixing, stiffeners, supports etc. to be Architect's satisfaction. Special attention shall be drawn to the following:-

    i. Wall hung cabinets shall be securely fixed to the wall with anchors or other fixing devices of adequate strength (eg. red head bolt) to cater for the dead and live loads of the cabinets. **Such loading shall not be less than 120 kg per 800mm wall hung unit**. A loading test will be performed on the wall hung cabinets upon request by the Architect. The fixing details shall be submitted for review prior to installation.

    ii. Composite Sinks shall be securely fixed to the sink cabinets. Stiffenings and supports shall be provided underneath drain boards of strength adequate to resist continuous impact, and to reduce the noise, due to the chopping of food on top of the drain boards.

    iii. To prevent sagging under load, intermediate supports shall be provided for the worktops and internal shelves inside any wall or base cabinet unit with a length exceeding 0.76m.

i. All hinges shall be strong, durable resistant to corrosion and be concealed.

j. The Contractor shall post a competent full time kitchen cabinet supervisor on site for the whole of installation period, and to supervise & monitor the installation.

k. The Contractor shall be responsible for the protection of his work including s.s. sink and mixer tap from damage after installation until substantial completion, i.e. handing over to the Employer and shall clean and touch up his completed works up to the Architect's satisfaction before handing over.

l. During installation, the Contractor shall preserve and handle with care all existing adjacent work of other trades and shall be responsible to make good and pay all damage incurred by his installation.

m. Generally

    i. The timber shall be cut to the required sizes and lengths as soon as practicable after the works are begun and stored dry under cover in such a manner that air will circulate freely around it.

    ii. The whole of the work shall be framed and finished in a proper and workmanlike manner in accordance with the detailed drawings, and fitted with all necessary mild steel ties, straps, bolts, screws and all other fixings.

    iii. Joinery work shall be prepared immediately after the placing of the Contract and framed up, but not wedged, glued or in any other way permanently joined, and then stored. When fixing in positions is required, it shall be properly and permanently fixed and bonded together

iv. All joinery works shall be properly framed together, tenoned, shouldered, wedged, pinned, dovetailed and bradded as required. Any parts that warp or develop shakes or other defects shall be replaced before wedging up.
v. External joinery work shall be famed up, fixed together with approved adhesive and pinned with hardwood pins.
vi. Internal joinery work shall be cross-tongued and glued; the tongues shall be cut at right angles or diagonally to the grain of the wood.
vii. Joinery work unless otherwise specified shall be fixed with screws of sufficient length for the various purposes; fixing on face work shall be done with round lost head nails with heads neatly punched in.
viii. All joinery works unless otherwise specified shall be wrought and rubbed down coarse and fine glass paper to a perfectly smooth and even surface.
ix. Before starting fabrication of repetitive fittings, prepare prototypes and obtain approval before starting fabrication.

n. Dimensions

Dimensions of sections shown on the drawings are finished sizes. Allow for planning and sanding faces to finished sizes. Check site dimensions before prefabricating joinery fittings.

o. Unexposed Surfaces

Treat the following unexposed surfaces with one coat of wood preservative :-

i. Backs of door frames,
ii. Backs of skirting,
iii. Backs of cupboard and closet frames,
iv. Backing fillets, etc.

p. Exposed Surfaces

All exposed surfaces, unless otherwise shown on drawings, shall be finished smooth using scrapes and multiple applications of glass paper before fixing.

q. Nailing

Nail timber sections securely to the backing and ensure that the nails do not split the timber. Split timbers shall be removed and replaced.

Punch nail heads below timber surfaces visible in completed work with approved sealer finished flush with face.

Nail weather boarding to wood framing with not less than two corrosion proofed nails in the width of each board at each framing member.

r. Screwing

When specified, screw timber sections to the backing including drilling pilot holes and countersinking heads flush with timber surfaces.

Screws shall be inserted full depth with a screwdriver and not hammered.

Countersink screw heads 5mm (minimum) below timber surfaces shall be left with natural finish. Glue in colour and grain matched pellets cut from matching timber. Finish off flush with face.

s. Wood preservative

Apply wood preservative as described in “painting” to all unexposed surfaces of timber. (e.g. Fake ceilings, backing fillets, back of door frames, cupboard framing, etc.

t. Hardwood Subframe

Whether or not mentioned on drawings, provide hardwood sub-frame of 25mm thick to the whole length of the back of the following fixtures :-

i. Plumbing fixtures built into walls, such as cabinets, metal soap and paper holders, etc.

ii. Tubular aluminium door frames,
iii. Recessed lighted fittings,
iv. Other metal fixtures built into walls

u. Skirting
Provide and fix hardwood or maple skirtings, where shown in the Schedule of Finishes or on drawings, on hardwood grounds securely plugged to walls, scribed to floors and properly mitred at angles.
All grounds behind skirtings shall be flushed up with plaster before skirtings are fixed to prevent voids behind the skirtings.

v. Window Cills and Pelmets
Window cills and pelmets shall be constructed according to drawings and Standard Details.
Window cills and pelmets shall be provided at locations as shown on drawings.

w. Plugging and Fastening System
Plugs for screws fixing to background of hard materials are to be proprietary plugs of plastic, soft metal, fibre or similar. The use of wood plugs shall not be permitted.
Power operated fastening system shall only be used in lieu of plugging when approved by the Architect.
The Contractor shall be responsible for compliance with all local regulations regarding the purchase, storage and safety measures to be undertaken and B.S. 4078 : Part 1 : 1987 code of practice for safe use of such systems.
Fastening system shall be used strictly in accordance with manufacturer's recommendations regarding the type and size of fastener suitable for the work.

x. Fixing Plasterboards
Plaster boards shall be fixed to timber bearers with galvanized nails :-
i. At 150mm centres, and
ii. Not closer than 15mm from edge or 20mm from cut ends.
Plaster boards shall be spaced 3mm to 6mm apart at joints.
All plaster boards shall be fixed and cleaned at least 24 hours before the application of plaster.
No plaster board shall be wetted before plastering.

y. Joinery Works Generally
Joinery works shall be constructed strictly in accordance with drawings and Standard Details.
Faces of framed joints to be square and to be driven together to give a close and accurate fit.
Prepare and frame up joinery work with dry joints and store until required for fixing.
Prior to fixing, open up all joints, put together with wedges and approved adhesive.
Any sections that have warped or developed shakes or other defects shall be replaced.
Plane timber for joinery on all faces. Finish exposed faces to a fine glasspapered surfaces and around arises to 1m radius.
All lippings should be finished in factory. If lippings have to be done on site to suit actual site condition, the gluing, material and application method shall be subject to the Architect's confirmation of no adverse comment.
All hinges shall be strong, durable resistant to corrosion and be concealed.

z. Doors Generally

Unless otherwise specified, the doors shall be of the thickness shown on drawings.

Provide 12mm lipping to all edges of flush doors.

Lipping to meeting edges of folding doors and swinging doors shall be 25mm thick rebated or rounded as required. Lipping shall be same wood as the door frame and veneer.

Doors shall be hung in accordance with C.P. 151 : Part 1 : 1957.

Bottom edges of door shall be primed before hanging.

aa. Framed, ledged and braced doors

Construct framed, ledged and braced doors of 45mm (minimum) thickness with 115mm wide styles and top rail, 225 mm wide middle and bottom rails, and 100 mm wide braces. Fill in with vertical boarding as 8.53.

ab. Panelled doors

Construct hardwood panelled doors 40 mm (minimum) thick, with 100mm wide styles, top rail and muntins and 200mm wide middle and bottom rails. Flat panels to be 20mm thick. Groove rebute or leave open framing, as specified, for panels or glass.

ac. Hollow Core Flush Doors

Unless otherwise shown on drawings, the doors shall be constructed as follows :-

Hollow core flush doors shall be framed up with 75mm hardwood stiles, top, intermediate and bottom rails all properly mortised and tenoned together.

Provide 25mm hardwood egg-crate core at 150mm centres both ways.

The doors shall be blocked out with 75mm wide hardwood piece for lock fitment.

Doors shall be covered both sides with 5mm plywood of :-

i. Luan faced finish, or
ii. Teak faced finish as specified.

Provide 1.3mm (minimum) laminated plastic sheeting bonded to 5mm plywood glued on the inside face of lavatory or bathroom doors as specified.

ad. Solid Core Flush Doors

Unless otherwise shown on drawings, the doors shall be constructed as follows :-

Solid core flush doors shall be framed up with 75mm hardwood stiles, top, intermediate and bottom rails all properly mortised and tenoned together.

Provide 50 mm hardwood vertical filling pieces bonded together under pressure.

The doors shall be blocked out with 75 mm wide hardwood piece for lock fitment.

Doors shall be covered both sides with 5mm plywood of :-

i. Luan faced finish, or
ii. Teak faced finish as specified.

ae. Solid Doors

Unless otherwise shown on drawings, the doors shall be constructed as follows :-

Solid doors shall be framed up with 150 mm hardwood stiles, top, intermediate and bottom rails all properly mortised and tenoned together.

Provide 75 mm hardwood vertical tongued and grooved boarding, V-jointed on face side and cramped with four 10mm diameter horizontal mild steel road each 15mm shorter than door width and threaded and fitted with nuts and washers at both ends at locations as specified on drawings.

All bolt holes shall be pelleted with hardwood of matching grain.

af. Openings in Doors

Frame opening with 12mm teak lipping all round.

Provide 12mm teak glazing bead for glazed openings.

Lippings and glazing beads shall be mitred at angles and fixed with screws.

ag. Door Frame and Architraves

Door frames and architraves shall be constructed strictly in accordance with drawings and standard details.

Door frames and architraves shall be in one length between angles.

Door frames and architraves shall be constructed of :-

i. Teak or,
ii. San Cheong where shown on drawings.

Door frames shall be constructed with posts tenoned into heads and put together with wedges and approved adhesive.

Architraves shall be mitred at angles joints.

Door frames shall be fixed to background with holdfasts of not less than three (3) for each jamb and secured to floor with dowels.

ah. Cupboard doors/ Vanity Counter doors/ Doors at Counters

Construct cupboard doors as follows, as specified :-

Plywood or blockboard teak lipped on all edges faced with laminated plastic sheet or prepared for painting, or

Hollow core doors to be as detailed in drawings.

ai. Drawers

Construct drawers with 18mm thick teak, hardwood or plywood front, as specified, 15mm thick hardwood or plywood back and sides, as specified, dovetailed and framed together and 5mm thick plywood bottom housed on three sides. Set drawers to slide on or with 15 x 15 mm hardwood screwed runners.

aj. Defects in Joiners Work

Should joints in Joiner’s work open or other defects, such as shrinkage, warping, winding arise within the Defects Liability Period stated in the Contract the cause thereof shall be deemed by the Architect to be due to unseasoned timber or faulty workmanship. Such defective joinery shall be taken down, rectified, redecorated or replaced if necessary and any work distributed shall be made good, all at the Contractor’s expense.

1.07 SUBMITTALS

A. Detailed Programme

a. The Contractor shall submit a fully detailed programme showing the various stages of works which include :-

i. Submission of shop drawings and the expected approval date;
ii. Samples submission;
iii. Fabrication;
iv. Delivery;

v. Installation;
vi. Cleaning;
vii. All other pertinent information.

b. The delivery and installation schedule shall suit the master programme, or such revised programmes as may be adjusted by the contractor from time to time to suit the site conditions. The delivery programme shall take into account the traffic and site storage conditions. Contractor shall undertake to provide and install at their own cost such provisional units / fittings as required in case of late delivery or loss by theft and/or vandalism in order to meet inspection schedules by appropriate authorities for the issuance of water certificate and Building Occupation Permit. Such provisional items shall be replaced by the original specified items at contractor's own cost when replacement is available. In any case, it is required that the installation of sinks and sink mixers shall be completed not later than two weeks prior to inspection by Water Authority in connection with the issuance of water certificate or inspection by B.D. in connection with the issuance of Building Occupation Permit whichever is the earlier.

c. The erection programme shall include detail description of the procedure for :-
i. Hoisting;
ii. Fixing;
iii. Protection; and
iv. Cleaning.

B. Manufacturer's Literature

The Contractor shall submit the Manufacturer's Installation Specification of all pantry cabinets, work counters, information counters and all built-in furniture / bookshelves / carpentries & other accessories for the Architect's approval.

C. Shop Drawings

a. The Contractor shall submit detail & complete shop drawings for all types of pantry cabinets, work counters, information counters and all built-in furniture / bookshelves / carpentries & other accessories indicating all relevant dimensions & details prior to placing order of materials.

b. The shop drawings shall consist of complete scaled plan & elevation drawings and details of all parts of the kitchen and pantry cabinets. Elevations & Sections shall be drawn in minimum 1:20 scale and all blow-up details shall be in full size.

c. Necessary associated works required to be done by other trades must also be indicated in the shop drawings.

d. It is the responsibility of the Contractor to verify all dimensions on site and to incorporate the verified dimension into the final set of shop drawings before manufacturing. No claim shall be entertained against discrepancies between as-built and design dimensions.

e. Five (5) printed copies and one (1) reproducible print for each of the final approval shop drawings shall be submitted for the Architect's use.

f. The Architect may reject, comment or approve such shop drawings. The Contractor shall re-design, and amend his shop drawings in compliance with the Architect's comments for approval.

g. No fabrication shall commence until shop drawings had been confirmed by the Architect with no adverse comment. No claim will be accepted for disapproval or amendments by the Architect. The comments by the Architect of any such shop drawings shall not relieve the Contractor of his duties and responsibilities under this contract.

h. Should the Contractor consider it necessary to fabricate the materials before comment in order to meet completion date, it is entirely at his own risk and no compensation in time and cost will be granted if the material and / or workmanship is eventually not acceptable to the Architect.

i. Interface

The Contractor shall be submit the following drawings indicating the following required location of works for Architect's approval prior to actual carrying our of works :

i. Location of hot and cold water supply pipes and drain pipes in relation to the units;

ii. Location of power points / lighting switches / MCB in relation to the units;

iii. Location of LPG/gas meter and pipes in relation to the units.

D. Samples

a. The Contractor shall submit a schedule of submission, to the Architect for confirmation of no adverse comment. The Contractor shall submit one (1) set of samples of cabinet components, including ironmongeries, for the Architect's confirmation of no adverse comment prior to placing order of materials.

b. Transparent finish for each species of wood veneer laminated to MDF or particleboard core, 300mm x 300mm., for each finish specified or shown consisting of veneer pieces cut from selected flitch samples, laminated to panel product, for each species and cut. Include at least one face-veneer seam and finish one-half of face as specified. Range is to constitute five number samples, and is to represent the maximum range in finish of the timber veneers.

c. Each finish type of high-density stratified natural timber panel, 600mm wide x 900mm high.

d. Submit chip sample of each type, finish, and colour of plastic laminate. Plastic laminate colour will be selected by the Architect.

e. Samples of curved pieces; scarf and interlocking finger jointing; bullnose, reveal, and edge treatments and fixing hardwares shall be submitted in 300mm x 300mm size or length to the Architect for review and approval.

E. Fire-retardant and preservative treatment certificates. Mill certificates for all timber and board materials.

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY, STORAGE AND HANDLING

A. All apparatus, materials, components and accessories shall be:-

a. Delivered to the site in a new condition;

b. Properly packed and protected against damage due to handling, adverse weather or other circumstances;

c. As far as practicable, they shall be kept in the packing cases and under protective coverings, tapes or coatings until required for use.

B. Any item suffering damage in transit or on the site shall be rejected and replaced without extra cost to the Employer and no item so rejected will be considered as a reason for extension of time.

C. In the case of components and accessories which originate outside Hong Kong:

a. All items of components and accessories shall be adequately and securely packed for safe transportation with due regard to the climatic conditions encountered in transit and on arrival;

b. The Contractor shall, at the time of shipping each consignment of components and accessories provide to the Architect in triplicate, packing lists and Bills of Lading which shall contain full statements of the packages consigned, with particulars of the dimensions, weights, contents, shipping marks and approximate value of each package;

c. BS 1133 and Supplements, or other comparable and acceptable code, shall be used as a guide for the standard of packing and package required. All bright, polished or plated parts shall be treated with suitable rust inhibitor.

D. Laminate should be stored horizontally with the top sheet turned face down and a cane board placed on top to protect the material from possible damage and reduce the change of new page of the top sheets. Laminate should be protected from moisture and should never be stored in contact with the floor or outside wall.

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

A. Wall hung cabinets shall be securely fixed to the wall with anchors or other fixing devices of adequate strength (eg. red head bolt) to cater for the dead and live loads of the cabinets. The Contractor shall carry out loading test to samples of the wall cabinets & drawers, upon request by the Architect, by applying the following storage load to the test samples :

a. Wall cabinets: 120 kg per 800mm

b. Drawers : 15kg

B. The load shall be maintained for a period of 24 hours. During the test there shall be no temporary or permanent distortion, weakening of the joints and anchorage points or failure of supports caused to the test sample.

1.12 WARRANTY

This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 - PRODUCTS

2.01 APPROVED MANUFACTURER

Subject to compliance with the Project Contract Documents, provide architectural woodworks and required accessories from the following approved manufacturers/ suppliers, or equivalent approved by the Architect:

a. Formica (Asia) Ltd
   Room 1305-9, 13/F, Olympia Plaza,
   255 King's Road, North Point, Hong Kong
   Tel: 2169 8111

2.02 MATERIALS

A. Refer also ASD GS 2012 clauses 13.12 to 13.21 & 13.24 to 13.30, and 21.30 to 21.37.

B. Supply and install materials according to the following:

a. Door - Melamine faced chipboard with PVC lipping edge.

b. Carcase shall be minimum 19mm thick 650-680 kilograms per m^3 density, faced with melamine 100gms per sq. metre with the following:

   i. Tested for screw holding, surface soundness tensile strength to BS5669. Strength and performance of complete units to level 3 of BS4875 Part 7.

   ii. Construction shall be of flat pack with cam & dowel fixings. Screw holding and surface soundness tensile strength shall be tested to BS5669. Strength and performance of complete units to level 3 of BS4875 Part 7. All surfaces are faced on both sides with minimum 1.2mm (minimum) thick permanently bonded (heated and pressured) melamine films. The surfaces shall be highly durable, anti-static, and resistant to heat (to approximate 130ºC) as well as to all household solvents. Colour shall be of matching colour subject to Architect's approval.

   iii. Front edges of carcase shall be melamine or polyester resin minimum 0.6mm thick of matching colour to that of the door front and the shelves subject to Architect's approval.

   iv. All exposed side panels shall match the colour of door panel with gloss/matt finish to be approved by Architect.

   v. All carcase edge shall be covered for protection against moisture.

   vi. Shelves flush with front (32m deep or as shown in Drawings) shall be in white laminate with 19mm thick sides and bases. Front edges shall be rounded and finished with PVC trim.

   vii. Sink base units shall be supplied with moulded plastic protective base covers for protection of water dripping.

c. Back Panel - Shall be of high-density decorative hardboard of minimum 3.2mm thick. The rear panels shall be securely slotted into the cabinet sides and the base and shall match the colour of carcase in above section 5.3b.

d. Adjustable Shelves

i. Adjustable shelves shall be made of minimum 19mm thick multi-layered chipboard covered on both sides and rounded with melamine resin to DIN 68765. The covered drill holes shall be at spacing not more than 32mm in the carcase sides to allow for height adjustment.

ii. The shelves shall be supported by secured and strong metal shelf pegs allowing minimum load-bearing capacity 50kg per m² of shelf surface. Ingenious shelf support clip shall be provided to prevent accidental vertical and transverse movement of the shelves.

iii. Base unit shelves shall be adjustable to 2 heights. Wall unit shelves shall be adjustable to 3 heights.

e. Drawer System (Read this in conjunction with ASD GS 2012 clause 13.67)

i. Side Panels - White metal sided steel drawer or aluminium sided drawer system (to Architect's approval) with roller action (self-closing) with fully adjustable front. The drawer base and back panels shall be made of 16mm thick chipboard laminated with melamine on both sides.

ii. Drawer Depth : 500mm minimum

iii. Drawer Height : 120mm minimum

iv. Drawer and Pull-out Elements

- White metal sided steel drawer system with roller action (self-closing) with fully adjustable front. The system shall be long-lift, maintenance-free, concealed runners, fully extendible, high lateral stability, smooth running, self-closing.
- The base and back of the drawer are of 16mm white Melamine Faced Chipboard.
- Drawers and pull-out base bottom shall have non-slip coating to prevent sliding of objects while removing and pulling-out the drawers.

v. Load Capacity

Minimum 30kg for width up to 600mm. The flexible rail system shall have additional divisions and system dividers to allow for maximum use of storage area.

vi. All drawer pull-outs in base and tall units shall glide along tandem with full pull-out runners with horizontal and vertical roller bearings and shall have a load-bearing capacity of 30kg. Automatic self closure shall take place at 60mm before the final closed position. Bottom with non-slip base material.

vii. The drawer front panels of the pull-outs can be adjusted in 3 dimensions.

f. Hinges

Unit doors shall be installed with concealed self-closing non-corrosive metal hinges fully adjustable in 3 directions (including height adjustment) with opening angle of 105º. Adjustment screws shall be used on the hinges to allow for perfect alignment of doors in 3 directions.

g. Base Cabinets

All wall units shall be fitted with solid, 3-dimensionally adjustable hanging brackets. The bracket shall not be visible inside the unit. The load-bearing capacity per bracket shall be in accordance with DIN 68930 100kg. The bottom carcase shelves shall be fitted with a wall spacing adjustment facility.

h. Corner Base Cabinets

i. The corner base cabinet is supplied with a blanking panel and fillet extension in 16mm MFC to match the door finish. This arrangement provides great flexibility in the positioning of the corner base relative to the adjacent cabinet.
ii. The blanking panel is drilled half depth from the back for an approved standard corner fillet offset, other offsets can be drilled on site. Appropriate fillet extension shall be available.
iii. Blanking panels are supplied and secured on site with four dummy drawer brackets which are screwed into the base and underside of the top rail.
iv. All parts and fittings are supplied shrink wrapped as a corner fillet kit.

i. Corner L Cabinets
   i. The cabinet has a 16mm MFC top, base and angled back. The sides and angled back are grooved to accept the 3.2mm white finished back boards which are pinned to the rear edges of the top and bottom panels.
   ii. The cabinet is supported on 5 adjustable legs, has a bi-fold door and the option of shelves or carousel.
   iii. Top and base panels are edged white on the facing edge. Shelf is edged white on the facing edge.

j. Wall Cabinets
   i. Sides to be 727mm high (or as shown in Drawings) and to be non-handed (invert for opposite hand) and edged all round with white PVC.
   ii. The cabinet base to flush allowing for light pelmet. Base and tops to be edged white on the front and reverse edges.
   iii. Tops and bottoms to be located by four 8mm dowel and cam fixings at each end.
   iv. 600 medium wall corner cabinets to be fitted with similar vertical rails as the base cabinet in white.
   v. Backs are 3.2mm white board and to be pinned to the back edges of the tops and bottoms during assembly.
   vi. One adjustable shelf is supplied on medium wall cabinets. Shelves are edged all round with white PVC.
   vii. Cabinets are hung using the adjustable hanging bracket.

k. Corner Wall Cabinets
   i. Corner wall cabinets have a 16mm carcase colour matched MFC blanking panel attached to the carcase with four brackets which locate in the hinge plate holes in the side and in the vertical rail.
   ii. Vertical rails are common to the double base cabinet.
   iii. Carcase edging is as other wall cabinets.

l. The Storage Cabinet
   i. The cabinet has three fixed horizontal panels (top, base and intermediate shelf) secured to the sides with six 8mm dowels and cam fixings at each end. It has one adjustable shelf for the 565 compartment and two for the 1392 compartment. Shelves are 358mm deep and edged in white PVC.
   ii. The one piece back panel is of 3.2mm board slotting into grooves and pinned to the top, mid and bottom panels to form a rigid box.
   iii. There are two back rails to give additional rigidity, the upper rail being accessible through two holes in the back panel to enable the cabinet to be fixed to the wall.

m. Tall Oven/Fridge Housings
Oven and fridge housings correspond to the existing range in terms of accommodation; door/drawer/pan drawer arrangements etc. and have similar adjustment facilities.

n. Light Pelmet
Light pelmet shall be 50mm in height chipboard with melamine resin finish, colour and pattern to be submitted for Architect's confirmation of no adverse comment. Profile of light pelmet shall be subjected to the final Architect's confirmation of no adverse comment.

o. Cornice
Cornice shall be chipboard with melamine resin finish, colour and pattern to be commented by Architect. Profile of cornice shall be subjected to the final Architect's confirmation of no adverse comment.

p. Overshelves with Spotlights
Overshelves shall be provided for wall units tall units and shelves as per drawing make of 19mm thick melamine faced chipboard in high gloss finish. Profile of overshelves shall be subjected to the final approval of the Architect.
Low voltage spot lights to be supply and installed on overshelves as per drawing and schedule and in accordance with respective supplier's technical specification.

q. Handles
Unless otherwise shown in Drawings or required and approved by Architect, D-shaped bow handle (or equivalent) to the final confirmation of no adverse comment from the Architect.

r. Plinth
Plinth screens are 16mm thick and 100mm high. An integrated base seal to prevent moisture penetration. Surface finish is in door matching colour in carcase material.

s. Thickened Floor
For sitting area, the contractor shall ensure the construction of thickened floor could tight-fitty receive all the kitchen cabinets above.

t. Wall Mounts
Semi-recessed hanger with metal fitting in plastic housing adjustable in height by 15mm, in depth by 20mm. Load capacity per cabinet shall be 120kg minimum.

u. Solid Surfacing Material Worktop
13mm thick solid surfacing work top with 39mm bullnose front edge and 50mm high (cove-joint) backsplash and with 26mm thick wooden frame for worktop support.

v. Sink: Stainless steel sink complete with waste fitting, chain and plug.

w. Mixer Trap
Desk mounted single hole sink mixer with single lever handle and swivel outlet of colour to match the sink and to final Architect's confirmation of no adverse comment.

x. Bottle Trap
Supply and install 38mm white plastic bottle trap and connect trap to sink and existing drainage pipes.

y. Seal
   i. Junctions between pantry cabinet and wall, floor, plinth, etc. to be sealed by silicone sealant to keep free of water and pests.
   ii. Material and Colour of sealant shall be submitted to the Architect for confirmation of no adverse comment.

iii. Junction between sink and pantry unit to be watertight and sealed with the sealant.
iv. Backsplash should be provided to the junctions between all worktops and wall. The junctions between the backsplash and wall should also be sealed by the sealant.

z. The Contractor is required to supply and install compact single batten luminaries (fluorescent tubes) with high efficiency 26mm lamps packed in point-of-scale carton at and all other associated fittings for cabinet pelmet lighting. Length of luminaries to suit cabinet size.

C. Hardwood (H.W.)

a. Hardwood shall be certified as originating from a sustained resource or managed plantation.
b. Hardwood for carpentry or joinery shall be best quality “San Cheong” (Kapore), or approved equivalent.
c. Density of hardwood shall not be less than 720kg per cubic metre at 15% moisture content.
d. Moisture content in hardwood is to be measured in accordance with the methods stated in BS EN 1313-2:1999.

D. Softwood

Softwood for carpentry shall be Pine, Cedar, Spruce or China Fir or other species approved by the Architect. All timber shall be appropriately stamped or marked to identify origin and grade. All timber would be kiln dried and vacuum impregnated to New Zealand Standard H3, or equivalent, with Copper Chrome Arsenate, or as directed otherwise by the Architect.

E. Teak (Yau Muk)

a. Density of teak shall not be less than 650kg per cubic metre at 15% moisture content.
b. Teak shall be equal in every respect to Indian Selected Grade II.
c. At any cross-section, the number of growth rings to 25mm shall not be less than 8 and the slope of grain shall generally not be greater than 1/8.

F. Plywood

a. Plywood shall be one of the grades specified in BS 6566 and which, in the opinion of the Architect, is appropriate to the work as follows :
   i. Grade 1 — For use in its natural state. The veneer shall be free from knots, worm and beetle holes, glue stains or other defects.
   ii. Grade 2 — For use where subsequent painting or similar treatment is required. The veneer on the face side may have a few sound knots, occasional minor discolouration or stain and small inlay repairs.
   iii. Grade 3 — For use where the plywood is not normally visible. The veneer may have defects other than those specified above, provided that its serviceability is not affected.
b. Unless otherwise specified, thicknesses of plywood in joinery work shall be as follows :-

i.	Hollow core doors	:	5mm
ii.	Solid core doors	:	5mm
iii.	Panelling	:	5mm
iv.	Closet doors	:	5mm
v.	Cupboards doors	:	12 or 20mm

vi. Counter top : 20mm

c. Plywood for exposed use shall have one of the following face veneers on one or both sides as required :-
   i. Luan,
   ii. Teak,
   iii. Walnut,
   iv. White maple,
   v. Ebony
d. Plywood for unexposed use shall have Grade 2 veneer and bond with Type M.R. (Moisture Resistant and Moderately Weather Resistant) adhesive to BS 1204: 1993).
e. Unless otherwise specified, plywood for internal use shall be Grade 2 with Type 2 adhesive and plywood for external use shall be Grade 2 with Type 1 adhesive, the veneers being rotary cut Luan.
f. Generally the bonding adhesive between veneers shall be resin adhesive classified as moisture and weather resistant (M.R.) to BS 1203: 2001.

G. Veneered Facings
   a. Veneers shall be not less than 1.5mm thick and straight grained and properly bonded to the backing.
   b. Veneers of adjacent panels, doors etc. shall be matched.

H. Laminated Plastic
   a. Shall be 1.3mm (minimum) thick of selected colour and pattern
   b. Shall be glued with adhesive to B.S. 1204 : Part 1 and 2 : or recommended by the manufacturer in approved brand
   c. Shall be used strictly in accordance with manufacturer's instructions
   d. The backing core for postforming laminated plastic shall have an even, smooth and continuous surface.
   e. Laminated plastic shall be postformed and glued to the backing core strictly in accordance with the Manufacturer's instructions.
   f. Samples of the backing core and finished product shall be submitted to the Architect for approval prior to bulk fabrication.

I. PVC or Acrylic Sheet

PVC or acrylic sheet shall be clear, translucent or coloured, as specified, and to be approved by the Architect.

J. Acrylic Plastic
   a. Acrylic plastic shall be 'Perspex' manufactured by I.C.I. (Imperial Chemical Industries Ltd.) or other approved equivalent.
   b. Acrylic plastic shall be cut and installed strictly in accordance with manufacturer's instructions.

K. Nails and Screws
   a. Steel nails and copper nails shall conform to B.S. 1202: Part 1 and Part 2: respectively.
   b. Nails shall be wire type of mild steel or cut type of black rolled steel with 'bright' finish, unless otherwise specified.
   c. Screws shall conform to BS 1210: with countersunk heads, unless otherwise specified.
   d. Stainless steel, nickel copper alloy and brass screws with matching colour shall be used in general.

e. The use of steel screws will only be permitted for use in temporary works and sub-frames or timber furrings of suspended ceiling.
f. Nail lengths shall be not more than the total thickness of sections to be joined less 5mm, or not less than twice the thickness of section through which nails are driven.
g. Where the thickness of the outer section through which nails are being driven is less than half that of the section to which nailing is being done, the depth of penetration of the nails into the latter shall be not less than 10 diameters of the nails being used.
h. Screw lengths to be not more than the total thickness of sections to be jointed, less 5mm, or not less than one and a half times the thickness of section through which screw are driven.
i. Where the thickness of the outer section being screwed is less than half that of the section to which screwing is being done, the depth of penetration on the screwing into the latter to be not less than the thickness of the outer section.
j. Screw cups to be brass cups or stainless steel to BS 1491, Part 2 - Table 13 surface pattern.
k. All screws used for securing access panels etc. are to be countersunk chromium plated brass screws fitted into chromium plated brass cup securely bedded into the timber.

L. Plugs
a. Plugs for fixing to hard materials shall be proprietary plugs of plastic, soft metal, fibre or similar.
b. Fixing to friable materials, plasterboard and the like shall be proprietary fixings specially designed for that situation.
c. The use of wood plugs shall not be permitted.

M. Adhesives
a. Adhesive for wood shall be cold setting synthetic resin adhesive of :-
   i. Moisture-resistant and moderately weather-resistant (M.R.) type to B.S. 1204 : Part 1 for internal use;
   ii. Weatherproof and boil-proof (W.B.P.) type to B.S. 1204: Part 1 for external use or internal use under very damp conditions.
   iii. Cold setting casein glue to B.S. 1444
b. Adhesives for plywood shall be one of the following types :-
   i. Type 1 Phenol formaldehyde resin adhesive classified as weather and boil proof (W.B.P.) in B.S. 1203: 1979.
   ii. Type 2 Urea formaldehyde resin adhesive classified as moisture and weather resistant (M.R.).
c. Adhesive for fixing laminated plastic sheet shall be synthetic resin adhesive classified as weatherproof and boil-proof (W.B.P.) in B.S. 1204: Part 1 and 2.
d. Where the temperature exceeds 25 degree C, a “warm-setting” grade of adhesive shall be used.
e. The use of animal glues defined in B.S. 745 will not be permitted.
f. All adhesive shall be fresh and applied in strictly accordance with the manufacturer’s instructions.

N. Wood Preservatives
Shall be an approved proprietary brand exterior grade where completely concealed or not decorated and colourless, coloured or suitable for overpainting where likely to be exposed or in contact with painted finish.

O. Approval

Obtain approval before using explosive cartridge operated fixings. All fixings shall be in accordance with the Factories and Industrial Undertakings (Cartridge - Operated fixing tools) Regulations using tools, normally of the indirect acting type, plus pins and cartridges which correspond with the manufacturer's specifications for that tool. A tool shall only be used by a person holding a certificate of competency specifying the maker and model of the tool on which he has been successfully trained.

PART 3 – EXECUTION

3.01 EXAMINATION / INSPECTION

A. Prior to the installation of architectural woodwork, and at the Contractor's direction, meet at the project site to review the material selections, substrate preparations, installation procedures, coordination with other trades, special details and conditions, standard of workmanship, and other pertinent topics related to the Work. The meeting shall include the Architect, the Contractor, architectural woodwork fabricator, architectural woodwork installer, various manufacturer's representatives, and representatives of other trades or subcontractors affected by the installation.

B. Examine the substrates, adjoining construction and conditions under which the Work is to be installed. Do not proceed with the Work until unsatisfactory conditions have been corrected.

C. Verify locations of concealed framing, blocking, reinforcements, and furring that support woodwork by accurate field measurements before being enclosed. Record measurements on final shop drawings.

D. Condition woodwork to average prevailing humidity conditions in installation areas before installing.

E. Back-prime surfaces and exposed edges of all Work under this Section scheduled for installation on exterior of Project Building.

F. Preparation for Finishing: Sand work smooth and set exposed nails and screws. Apply wood filler in all exposed nail and screw indentations.

G. Verify mechanical, electrical, and building items affecting this Section are in place, tested, approved, and ready to receive the Work of this Section.

H. Prime paint surfaces of items and assemblies in contact with cementitious materials.

3.02 PREPARATION

(NOT USED)

3.03 INSTALLATION / APPLICATION

A. Coordination: Coordinate architectural woodwork with the adjacent work of other sections. Provide items to be placed during the installation of other work at the proper time to avoid delays. Coordinate placement of such items, including inserts and anchors, accurately in relation to the final location of architectural woodwork.

B. General: Install architectural woodwork in accordance with AWI Section 1700, Premium Grade.

a. Coordinate installation with the work of other trades to ensure exact fit and perfect alignment. Verify dimensions before proceeding and obtain measurements at job site for work required to be accurately fitted to other construction. Tolerances for cabinet door swings and drawer clearances shall also be verified prior to fabrication, installation or ordering of cabinets.
b. Install work plumb, level, true and straight with no distortions. Provide shims as required.
c. Cutting, trimming, fitting and matching of prefinished work will not be permitted.
d. Where cutting is required, scribe to fit adjoining work so as not to damage finished surfaces.
e. Securely fasten architectural woodwork items to blocking with concealed fasteners only. Where surface nailing is required, countersink and fill flush with the woodwork so that the finished heads are undetectable.
f. Install materials utilizing materials and methods as recommended by manufacturer unless otherwise specified.
g. Standing and Running Trims : Install with minimum joints, using maximum lumber lengths possible. Cope at returns; mitre at corner. Use scarf joints where necessary for base skirting. Butt joints are not permitted.
h. Casework: Provide well-fitting and smooth operating doors and drawers. All cabinet door and drawers shall have nylon stoppers. Provide run on drawer slides.
i. All arises shall be slightly rounded, 1.5mm radius.
j. All timber that is to be exposed in the finished surfaces of joinery works shall be dressed on the appropriate faces, unless otherwise stated. Exposed end grain shall be finished smooth to a degree equal to the overall finish. Joinery work shall generally be finished to a fine glass papered surface unless otherwise stated.
k. All panel trim to have 45 degree mitre corners, all edges and corners to be slightly eased or rounded to minimum 2mm radius. All joints in skirting, cornices, trim must be cut and overlapped 45 degree (scarf joint), no butt joints permitted. All panel trim shall be of a single length of timber without joints. In no case shall trim lengths be pieced together of short (under 3500mm) lengths.
l. Plastic Laminates:
    i. Cut, fit and bond sheets to the surfaces specified, or as shown on the Drawings, with a specified type WBP adhesive recommended by the Manufacturer of the laminate, and neatly trim as required. Butt chamfer where returned around salient edges. Perform bullnoses where indicated in strict accordance with the limitations of the material as recommended by the Manufacturer. Where radius corners and bullnosed edges are detailed, these shall be formed in post formed quality laminates.
    ii. All inside faces of laminated cupboards, built-in furniture etc. shall be fully laminated internally with matte white laminate, or as otherwise indicated on the Drawings.
m. Provide adequate fresh air, exhausting and other measures to ensure that specified indoor air quality during and after Work of this Section is met. All formaldehyde, volatile organic compounds and other toxics released during the Work of this Section shall be purged in strict accordance with the Consulting Mechanical Engineer's Contract Documents.

C. Panelling: Install panelling in accordance with AWI Section 1700 and as follows:-

a. Provide a system of concealed panel hanger clips and corresponding wall clips to support the panel system. Face nailing shall not be permitted.

b. Hang the panels in the designated locations. Scribe and cut panelwork to fit adjoining work, and refinish cut surfaces and repair damaged finish at cuts. Panels shall be straight, level, flat and flush with adjoining panels. Install to a tolerance of 1/8 in. in 96 in. for plumb and level. Install with no more than 1/16 in. in 96 in. vertical cup or bow and 1/8 in. in 96 in. horizontal variation from a true plane.

c. Where reveals are indicated, keep panels spaced so that reveals are parallel and of widths shown.

3.04 PROTECTION

A. Damage Prevention: Protect architectural woodwork so that it will be without damage at the time of acceptance.

a. During and after Finish Carpentry construction, all work subject to staining or damage shall be protected by heavy-duty boards. The protection shall be kept in position, and maintain in good condition, by the Contractor until Project Handover.

b. Protect all exposed corners and edges from foot and wheel traffic and impacts.

c. Protect all finished carpentry, casework, furnitures, and built-in items from damages, staining and adverse indoor psychometric conditions until Project Handover. The Contractor shall be liable for all repair, replacement and expenses for their failure in adhering to this requirement.

B. Repairing of Damage: Touch-up marred finishes to match adjacent surfaces perfectly.

a. Architectural woodwork which, in the opinion of the Architect, cannot be satisfactorily refinished in the field shall be removed and replaced, with units to match contiguous architectural woodwork in all respects.

C. Should any shrinkage or warping of casework, furniture, built-in items, etc. take place so as to impair the strength and/or appearance of the finished work during the Defect Liability Period, the Contractor shall make good such to the satisfaction of the Architect.

D. Defective joinery work shall be taken down, refitted, redecorated and/or replaced. Any Work disturbed shall be made good at the Contractor's expense.

3.05 CLEANING

A. Remove dirt, stain, finger prints, etc. upon completion of Finish Carpentry Work.

B. Clean soiled surfaces with cleaning solution compatible and approved by the Manufacturer of the finish coating.

C. Use non-metallic tools in cleaning operations.

D. Wax and re-polish all wood joinery as recommended by Manufacturer, prior to final handover.

END OF SECTION